King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Final Exam Term 241

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. The slope of the parametric curve $x = e^t,\ y = e^{-t}$ at $t = 1$ is

(a) $\frac{1}{e}$

(b) e^2

(c) $-\frac{1}{e}$

(d) e

(e) $-\frac{1}{e^2}$

2. $\lim_{(x,y)\to(0,0)} \frac{x^2+2xy^2+y^2}{x^2+y^2} =$

(a) 1

(b) 0

(c) 2

(d) -1

(e) DNE

3. The number of intersection points of the two polar curves $r = 1 - \cos\theta$ and $r = 1 + \sin\theta$ over $[0, 2\pi)$ is

(a) 2

(b) 3

(c) 1

(d) 0

(e) 4

4. The volume of the solid inside $x^2 + y^2 + z^2 = 16$ and outside $z = \sqrt{x^2 + y^2}$ is equal to

(a) $\dfrac{\pi}{3}$

(b) $\dfrac{32\pi}{3}\left(2 + \sqrt{2}\right)$

(c) $\dfrac{32\pi}{3}\left(1 + \sqrt{2}\right)$

(d) $\dfrac{64\pi}{3}\left(1 + \sqrt{2}\right)$

(e) $\dfrac{64\pi}{3}\left(2 + \sqrt{2}\right)$

5. $\int_0^{\sqrt{2}} \int_y^{\sqrt{4-y^2}} \sqrt{x^2 + y^2}\, dx\, dy =$

(a) $\dfrac{2\pi}{3}$

(b) $\dfrac{5\pi}{3}$

(c) $\dfrac{4\pi}{3}$

(d) $\dfrac{\pi}{2}$

(e) 2π

6. The x-coordinate of the point on the surface

$$z = 4x^2 + 4xy - 2y^2 + 8x - 5y - 4$$

at which the tangent plane is horizontal is

(a) $\dfrac{1}{12}$

(b) $-\dfrac{1}{3}$

(c) $-\dfrac{1}{4}$

(d) $-\dfrac{5}{12}$

(e) -1

7. Let T be the triangular region in the xy-plane with vertices $(2,0)$, $(0,1)$, and $(1,2)$ respectively. If M and m represent respectively the absolute maximum and absolute minimum of the function $f(x,y) = (2x-y)^2$ over the region T, then $M + m =$

(a) 19

(b) 17

(c) 14

(d) 18

(e) 16

8. The directional derivative of the function $f(x,y,z) = xy + yz + xz$ at the point $P(1,2,-1)$ in the direction of $\vec{v} = \langle 2,1,-1\rangle$ is

(a) $\dfrac{3}{\sqrt{6}}$

(b) $-\dfrac{1}{\sqrt{6}}$

(c) $-\dfrac{2}{\sqrt{6}}$

(d) $-\dfrac{5}{\sqrt{6}}$

(e) $\dfrac{2}{\sqrt{6}}$

9. The function $f(x,y) = \sqrt[3]{x^2+y^2} + 2$ has

(a) a relative maximum at $(0,0)$

(b) a critical point at $(0,1)$

(c) no extrema

(d) a relative minimum at $(0,0)$

(e) a saddle point at $(0,0)$

10. The volume of the solid bounded by $z = 4 - x - y$, $z = 0$, $y = 2$, $y = x$, and $x = 0$ is equal to

(a) 4

(b) 3

(c) 1

(d) 2

(e) 5

11. A normal vector to the level curve of $f(x, y) = y - \sin x$ at the point $(\pi, 0)$ is

(a) $\langle -1, 2 \rangle$

(b) $\langle -1, 1 \rangle$

(c) $\langle 1, 2 \rangle$

(d) $\langle 2, 1 \rangle$

(e) $\langle 1, 1 \rangle$

12. The minimum value of $f(x, y) = e^{-xy/4}$ subject to the constraint $x^2 + y^2 = 1$ is equal to

(a) 1

(b) $e^{\frac{1}{8}}$

(c) e^{-1}

(d) $e^{-\frac{1}{8}}$

(e) e

13. $\int_0^{\ln 4} \int_0^{\ln 3} e^{x+y} \, dy \, dx =$

(a) 1

(b) 4

(c) 6

(d) 3

(e) 5

14. An equation in spherical coordinates for the surface $x^2 + y^2 - 3z^2 = 0$ is

(a) $\phi = \dfrac{\pi}{3}$ or $\phi = \dfrac{2\pi}{3}$

(b) $\phi = \dfrac{\pi}{4}$ or $\phi = \dfrac{3\pi}{4}$

(c) $\phi = \dfrac{\pi}{4}$ or $\phi = \dfrac{\pi}{3}$

(d) $\phi = \dfrac{\pi}{2}$ or $\phi = \dfrac{3\pi}{2}$

(e) $\phi = \dfrac{\pi}{6}$ or $\phi = \dfrac{5\pi}{6}$

15. A vector orthogonal to both $\mathbf{u} = -3\mathbf{i} + 2\mathbf{j} - 5\mathbf{k}$ and $\mathbf{v} = \mathbf{i} - \mathbf{j} + 4\mathbf{k}$ is

(a) $-3\mathbf{i} - 7\mathbf{j} - \mathbf{k}$

(b) $3\mathbf{i} - 7\mathbf{j} - \mathbf{k}$

(c) $7\mathbf{i} + \mathbf{j} - 3\mathbf{k}$

(d) $\mathbf{i} + 7\mathbf{j} - 3\mathbf{k}$

(e) $3\mathbf{i} - \mathbf{j} - 3\mathbf{k}$

16. If

$$\int_0^4 \int_0^{(4-x)/2} \int_0^{(12-3x-6y)/4} dz\, dy\, dx = \int_0^a \int_0^b \int_0^c dy\, dx\, dz,$$

then

(a) $a = 3,\ b = \dfrac{12 - 4z - 3x}{6},\ c = \dfrac{12 - 4z}{3}$

(b) $a = 3,\ b = \dfrac{12 - 4z}{3},\ c = \dfrac{12 - 4z - 3x}{6}$

(c) $a = 12,\ b = 4,\ c = \dfrac{12 - 4z}{3}$

(d) $a = \dfrac{12 - 4z}{3},\ b = 3,\ c = \dfrac{12 - 4z - 3x}{6}$

(e) $a = \dfrac{12 - 4z}{3},\ b = \dfrac{12 - 4z - 3x}{6},\ c = 3$

17. A box without a top is to be made, where the material for the base costs twice as much per square unit as the material for the sides. If the sides cost \$4 per square unit and the total cost of materials is \$600, then the sum of length, width, and height of the largest possible box is:

(a) 15 units

(b) 20 units

(c) 35 units

(d) 30 units

(e) 60 units

18. $\int_0^1 \int_x^1 \cos(y^2)\, dy\, dx =$

(a) $\sin(1)$

(b) 0

(c) $\dfrac{1}{2}\cos(1)$

(d) $\cos(1)$

(e) $\dfrac{1}{2}\sin(1)$

19. If the volume of the solid region cut from the sphere $x^2 + y^2 + z^2 = 4$ by the cylinder $x^2 + y^2 - 2y = 0$ is given by

$$\int_0^{a\pi} \int_0^{b \sin\theta} \int_{-\sqrt{c-r^2}}^{\sqrt{c-r^2}} r \, dz \, dr \, d\theta,$$

then $a + b + c =$

(a) 5

(b) 10

(c) 8

(d) 6

(e) 7

20. Let (a, b, c) be the intersection point of the plane $5x + 3y = 17$ and the line $\frac{x-4}{2} = \frac{y+1}{-3} = \frac{z+2}{5}$. Then $a + b + c =$

(a) -3

(b) -2

(c) 1

(d) 0

(e) 4

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	$\frac{dy}{dx} = \frac{-e^{-t}}{e^t} = -e^{-2t}$, so at $t = 1$ it is $-\frac{1}{e^2}$.
2	**(a)**	$= 1 + \frac{2xy^2}{x^2+y^2}$ and $\left\lvert \frac{2xy^2}{x^2+y^2} \right\rvert \le 2\lvert x\rvert \to 0$, so the limit is 1.
3	**(b)**	$\tan\theta = -1$ gives 2 points, plus the pole (on both curves): 3 points.
4	**(e)**	Spherical, $\phi \in [\frac{\pi}{4}, \pi]$, $\rho \le 4$: $V = \frac{128\pi}{3}(1 + \frac{\sqrt{2}}{2}) = \frac{64\pi}{3}(2 + \sqrt{2})$.
5	**(a)**	Polar: $\theta \in [0, \frac{\pi}{4}]$, $r \in [0, 2]$: $\frac{\pi}{4} \cdot \frac{8}{3} = \frac{2\pi}{3}$.
6	**(c)**	$z_x = 8x + 4y + 8 = 0$, $z_y = 4x - 4y - 5 = 0$; adding gives $x = -\frac{1}{4}$.
7	**(e)**	$f \ge 0$ with $f(1, 2) = 0$, so $m = 0$; vertex values $16, 1, 0$ give $M = 16$; $M + m = 16$.
8	**(b)**	$\nabla f(P) = \langle 1, 0, 3\rangle$, so $D_{\mathbf{u}} f = \frac{2-3}{\sqrt{6}} = -\frac{1}{\sqrt{6}}$.
9	**(d)**	$f \ge 2 = f(0, 0)$ with equality only at the origin, so a relative minimum at $(0, 0)$.
10	**(a)**	$V = \int_0^2 \int_0^y (4 - x - y)\, dx\, dy = \int_0^2 (4y - \frac{3}{2}y^2)\, dy = 4$.
11	**(e)**	$\nabla f = \langle -\cos x, 1\rangle$, which at $(\pi, 0)$ is $\langle 1, 1\rangle$.
12	**(d)**	Minimize f by maximizing $xy = \frac{1}{2}$ on the circle: $f_{\min} = e^{-1/8}$.
13	**(c)**	$\int_0^{\ln 4} e^x\, dx \cdot \int_0^{\ln 3} e^y\, dy = (4 - 1)(3 - 1) = 6$.
14	**(a)**	$\rho^2 \sin^2\phi = 3\rho^2 \cos^2\phi \Rightarrow \tan^2\phi = 3$: $\phi = \frac{\pi}{3}, \frac{2\pi}{3}$.
15	**(a)**	$\mathbf{u} \times \mathbf{v} = 3\mathbf{i} + 7\mathbf{j} + \mathbf{k}$; its negative $-3\mathbf{i} - 7\mathbf{j} - \mathbf{k}$ works.
16	**(b)**	Plane $3x + 6y + 4z = 12$: $z \in [0, 3]$, $x \in [0, \frac{12-4z}{3}]$, $y \in [0, \frac{12-4z-3x}{6}]$.
17	**(a)**	Cost: $xy + xh + yh = 75$. Maximizing xyh gives $x = y = h = 5$, sum 15.
18	**(e)**	Swap order: $\int_0^1 y\cos(y^2)\, dy = \frac{1}{2}\sin(1)$.
19	**(e)**	Cylinder $r = 2\sin\theta$, $\theta \in [0, \pi]$; sphere $z = \pm\sqrt{4 - r^2}$: $1 + 2 + 4 = 7$.
20	**(c)**	Parametrize the line; the plane gives $t = 0$, point $(4, -1, -2)$, sum $= 1$.